

Economic Analysis of Text and other Non-Standard Data ISEG Summer School 2026

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11. Audio as Data

Outline

Introduction

Classical Machine Learning

Recurrent Neural Nets

Human Hearing is Interesting

- ▶ Hearing is a critical sense for communication and survival
- ▶ Like vision, it enables us to interact with the world without physical contact
- ▶ Various areas of the brain are associated with auditory processing
- ▶ Humans are "**analysts**" and "**generators**" of audio at the same time
- ▶ Human hearing has constraints:
 - ▶ Limited frequency range: 20 Hz to 20 kHz
 - ▶ Difficulty in locating low-frequency sounds
- ▶ Fascinating questions:
 - ▶ Fundamental: How does the auditory system decode sounds? (Beyond scope of this class)
 - ▶ Aggregate/social: What is the impact of sound on societal outcomes? (Central to our class)

Audio Analysis Definitions

- ▶ Many view it as a signal-processing challenge
- ▶ Julius O. Smith III: reconstructing auditory experiences
- ▶ This class explores techniques to analyze auditory data to study economics questions
- ▶ Focus on existing auditory data, not the mechanics of data capture

What is Sound?

- ▶ A sensory experience created by vibrations traveling through a medium (usually air)
- ▶ Analogy to ocean waves: sound waves propagate through air like ocean waves move through water
- ▶ As ocean waves interact with air, they create ripples in the atmosphere, analogous to **sound waves**

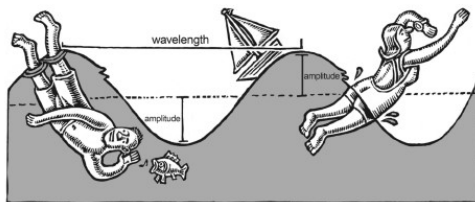


Figure: Ocean waves analogy to sound waves (Source: Sulzer, 2021)

Humans typically hear between 20 Hz and 20 kHz.

- ▶ We begin to hear low vibrations at around 20 Hz
- ▶ Teenagers can perceive frequencies up to about 20,000 Hz.
- ▶ Each octave doubles the frequency.
- ▶ Humans with good hearing can perceive a ten-octave range.

1. 20 Hz
2. 40 Hz ($20 \text{ Hz} \times 2$)
3. 80 Hz ($40 \text{ Hz} \times 2$)
4. 160 Hz ($80 \text{ Hz} \times 2$)
5. 320 Hz ($160 \text{ Hz} \times 2$)
6. 640 Hz ($320 \text{ Hz} \times 2$)
7. 1280 Hz ($640 \text{ Hz} \times 2$)
8. 2560 Hz ($1280 \text{ Hz} \times 2$)
9. 5120 Hz ($2560 \text{ Hz} \times 2$)
10. 10240 Hz ($5120 \text{ Hz} \times 2$)
11. 20480 Hz ($10240 \text{ Hz} \times 2$)

Figure: Human hearing range

Key Concepts: Hz, Octave, and Sampling Frequency

- ▶ **Hz (Hertz):** Measures the number of cycles per second of a periodic waveform
 - ▶ Higher Hertz means a higher-pitched sound
 - ▶ If a tone has a frequency of 440 Hz, the sound wave completes 440 cycles every second (A4)
- ▶ **Octave:** The interval between one musical pitch and another with half or double its frequency
 - ▶ 440 Hz and 880 Hz
- ▶ **Sampling frequency:** The number of samples per second from a continuous signal to make a digital signal:
 - ▶ Standard CD sampling rate is 44.1 kHz...
 - ▶ ...meaning that the audio waveform is being sampled 44,100 times per second

The wavelength is an important sound wave characteristic

- ▶ What is a sound wave, to begin with?
 - ▶ A transfer of sound energy through a medium
 - ▶ It is transmitted via the vibration of particles within that medium
- ▶ Illustration for non-physicists:
 - ▶ When something vibrates (like a speaker's diaphragm), it pushes on the neighboring air particles, increasing the pressure in that region → this push is then transferred from particle to particle → when it reaches someone's ear, the ear interprets the varying pressures as sound
- ▶ The physical distance between identical points in consecutive cycles of a sound wave
 - ▶ Determines the pitch

Frequency and Wavelength

The wavelength of a sound wave depends on its frequency.

$$\lambda = \frac{v}{f}$$

- ▶ λ = wavelength (distance between two wave peaks)
- ▶ v = speed of sound in air (343 m/s)
- ▶ f = frequency in Hertz

Implication

- ▶ Higher frequency $\rightarrow$ shorter wavelength
- ▶ Lower frequency $\rightarrow$ longer wavelength

Example:

- ▶ A tone of 440 Hz (concert A) has wavelength

$$\lambda = \frac{343}{440} \approx 0.78 \text{ meters}$$

The amplitude is another important characteristic

- ▶ Amplitude in a general wave context: the magnitude of change in the oscillating variable within the wave
 - ▶ → Larger amplitudes mean more energy and often translate to louder sounds when talking about sound waves
- ▶ Amplitude in digital audio signal processing : understood as instantaneous amplitudes that represent the “magnitude of change” in the pressure wave (the sound wave) from its equilibrium position at that particular moment
 - ▶ The specific value of the audio waveform at a given sample point

One-Tone Sound Wave

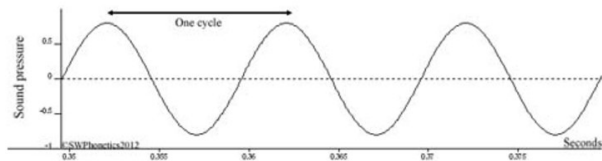


Figure: Sidney Wood and SWPhonetics, 1994-2024

A More Complex Sound Wave

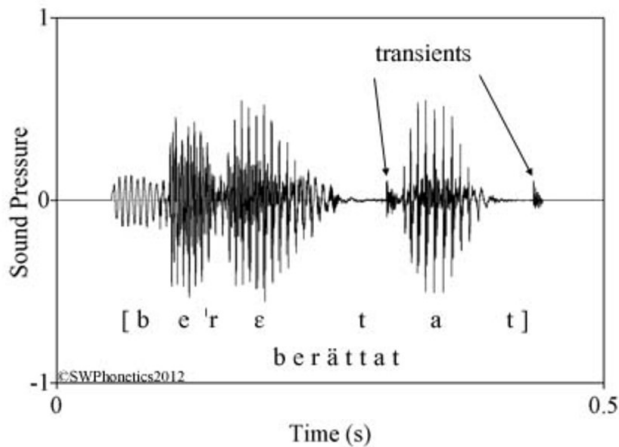


Figure: Sidney Wood and SWPhonetics, 1994-2024

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Recap from computer vision: For classical ML, we (often) extract features explicitly.

- Recall the typical pipeline for classical machine learning:

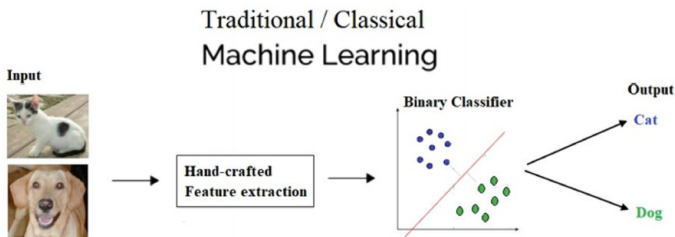


Figure: Dey (2018)¹

¹Dey, S. (2018). Hands-On Image Processing with Python: Expert Techniques for Advanced Image analysis and Effective Interpretation of Image Data. Packt Publishing Ltd.

We can use a similar workflow for audio data.

- Recall the typical pipeline for classical machine learning:

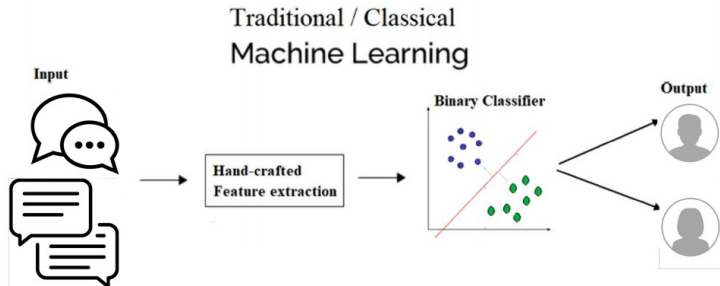


Figure: Adaptation by Widmer (2023) of Dey (2018)²

²Dey, S. (2018). Hands-On Image Processing with Python: Expert Techniques for Advanced Image analysis and Effective Interpretation of Image Data. Packt Publishing Ltd.

Can we use the raw audio input for classification tasks?

- ▶ Neural networks, such as CNN, can handle raw audio data → extract relevant features automatically (cf. images part)
- ▶ For very simple tasks, it may be possible with classical machine learning
- ▶ More generally, audio waveforms are high-dimensional data, especially for long clips
 - ▶ Computationally demanding
 - ▶ And often not necessary → suitable format depends on task
- ▶ For both classical and neural approaches, feature extraction is often useful

Spectral features capture important characteristics of a sound's frequency content

- ▶ Spectral features are widely used for audio tasks (e.g., music analysis, speech processing, audio classification)
- ▶ Typically, they encompass spectral centroid, spectral bandwidth, or spectral flatness
- ▶ The spectral centroid shows the frequency spectrum's “center of gravity”
 - ▶ Calculated as the weighted mean of the frequencies in the signal, with their amplitudes being the weights
 - ▶ Higher values reflect brighter sound
 - ▶ Technically, we cut the sound into frames and apply a Fourier Transform per frame
 - ▶ Hence, we transform the frame's time-domain signal into the frequency-domain

Spectral features beyond the spectral centroid

- ▶ The spectral bandwidth describes how wide the frequency band is
 - ▶ A wider bandwidth implies a broad range of frequencies
 - ▶ Conversely, a narrower bandwidth indicates a more tonally pure sound
 - ▶ It can help, for example, to identify the complexity of sound
- ▶ The spectral flatness indicates how “noise-like” a sound is instead of being tonal
 - ▶ Values close to 1 indicate a noise-like sound
 - ▶ Values near 0 indicate a more tonal sound
 - ▶ It can be helpful, for instance, to disentangle tones and environmental noises

Mel-Frequency Cepstral Coefficients (MFCCs) are often used for feature extraction

- ▶ MFCCs summarize the spectral structure of a sound in a way that reflects human hearing.
- ▶ Humans perceive sound frequencies non-linearly (rather, logarithmically)
- ▶ Human sensitivity is greater to changes in lower frequencies
- ▶ The Mel scale is a way to measure pitch that matches how we hear sounds
 - ▶ It is a perceptual scale of pitches judged to be equal in distance from one another
 - ▶ Originally derived from experiments with human listeners
- ▶ MFCCs represent the power spectrum of an audio signal more in line with human hearing
 - ▶ They use the Mel scale

MFCCs capture the short-term power spectrum of sound

- ▶ One typically begins by dividing the audio signal into short (overlapping) frames, for instance 20-40 ms
 - ▶ This allows assuming stationarity within each frame
 - ▶ Stationarity means that the statistical properties of the signal (like mean, variance) are constant over the frame's duration
- ▶ Several processing steps involved (including fast fourier transform)
- ▶ Bottom line: Typically end up with 12-13 coefficients per frame
- ▶ These coefficients capture the **overall spectral envelope of the sound**.
 - ▶ Empirically found to capture the most important features

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Recall that feature extraction is used for many audio (classification) tasks

- Recall the typical pipeline for classical machine learning

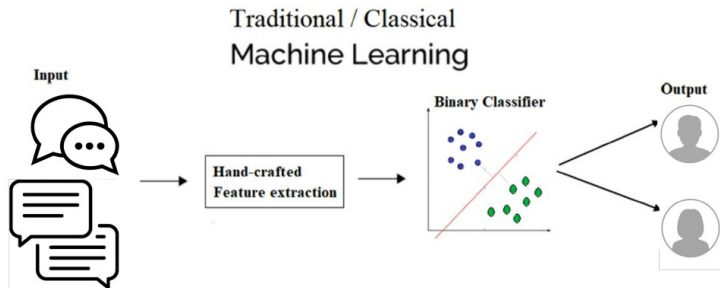


Figure: Adaptation by Widmer (2023) of Dey (2018)³

³Dey, S. (2018). Hands-On Image Processing with Python: Expert Techniques for Advanced Image analysis and Effective Interpretation of Image Data. Packt Publishing Ltd.

What models are used in audio analysis with deep learning?

- ▶ The audio analysis landscape is diverse
- ▶ CNN are used for classification with audio data
 - ▶ But less prominently than in CV
 - ▶ Audio data is often transformed into spectrograms (time-frequency representations) to use CNN
 - ▶ Examples: Emotions, speaker identity, accents
- ▶ Otherwise, varied, possibly hybrid approaches (RNN, transformers)
- ▶ In any case, audio data often requires pre-processing
- ▶ Other methods in audio analysis:
 - ▶ Classical ML techniques (e.g., MFCC)
 - ▶ Unsupervised and semi-supervised approaches: Useful with limited labelled data

Audio data is sometimes used with CNN or RNN

- ▶ Tailored for tasks with temporal dependencies
- ▶ For CNN, the conceptual foundation discussed in the computer vision part also applies here
- ▶ Recurrent Neural Networks (RNN) are “specialists” for sequential data
 - ▶ Sequential data examples are audio or text
 - ▶ RNN have a memory mechanism to remember previous inputs
 - ▶ They can, thus, “remember” context in sequences
 - ▶ For sequences, context is important to understand the overall pattern

Conceptually, how does an RNN proceed

- ▶ The data is typically pre-processed
 - ▶ Extract features: for example, MFCC by time t (often relatively short snippets)
 - ▶ Each feature at time t becomes input x_t to the RNN
- ▶ $\forall t \in \{0, 1, 2, \dots, T\}$, x_t are processed by the RNN sequentially
- ▶ RNN computes new hidden state h_t for each time step t
- ▶ Typically: $h_t = \text{Activation}(W_{xh} \cdot x_t + W_{hh} \cdot h_{t-1} + b)$
- ▶ W_{xh} , W_{hh} are weight matrices; b is a bias vector
- ▶ RNN learns W_{xh} , W_{hh} , and b
- ▶ Hidden state h_t used for output (e.g., in classification)